

Prathamesh Rane

Software Developer | Front-End Developer | Python Developer

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🐙 GitHub 🔗 LinkedIn

About Me

Master's student in Computational Modeling & Simulation at TU Dresden with experience in front-end development, Python programming, software development, databases, and AI/ML research. Skilled in React, Next.js, JavaScript, Python, SQL, Linux/Unix, Git, and structured problem solving. Currently developing a full-stack booking platform as an independent project and interested in contributing to real-world software, automation, testing, and digitalization projects in an international engineering environment.

Technical Skills

Programming Languages: Python, JavaScript, C, C++, Java, R

Front-End Development: React, Next.js, HTML, CSS, JavaScript, Bootstrap, responsive UI design

Databases & Data Formats: SQL, XML, PostgreSQL, MySQL, MS SQL, MongoDB

Development Tools: Git, GitHub, Linux/Unix, Shell basics, VS Code, Conda, Jupyter Notebook

Software Development: Object-Oriented Programming, debugging, testing basics, structured programming, automation, documentation

Machine Learning & Data: PyTorch, TensorFlow, NumPy, Pandas, OpenCV, Transformers, Matplotlib

Design Tools: Figma, Adobe XD, Canva, Adobe Photoshop

Education and Training

Master of Science - Computational Modeling and Simulation

10/2024 - Current

Technical University of Dresden

Dresden, Germany

- Relevant focus: software systems, data engineering, machine learning, simulation, logical modeling, and technical computing.

Bachelor of Engineering - Information Technology

07/2020 - 08/2024

University of Mumbai

Mumbai, India

- Scored 93% in the C Programming Examination conducted by IIT Bombay.
- Contributed to 5+ GitHub projects and actively worked on web and software development projects.
- Core member of Research Cell and Computer Society of India student chapter.
- Collaboratively developed a PHP-based web application: shahandanchor.com/research

Work Experience

Developer

08/2022 - 08/2023

Aviation Horizons Limited

Mumbai, India

- Created aircraft models and developed flight simulation components based on black box data.
- Worked on technical modeling, data interpretation, and simulation-based software workflows.
- Applied structured problem-solving and programming concepts to support technical software development.

Ops Associate

02/2025 - 04/2025

Flink

Germany

- Prepared customer orders, managed inventory, restocked products, and performed quality checks in a fast-paced logistics environment.
- Worked with digital inventory systems, structured operational processes, and time-sensitive team coordination.

Creative Co-Lead

11/2021 - 04/2022

SAKEC Virtual Lab Website

Mumbai, India

- Worked with front-end and graphic design teams to maintain and update website functionalities.

Simulation Developer

11/2021 - 04/2022

SAKEC Chemistry Virtual Lab

Mumbai, India

- Created chemistry models and animated experiments for a virtual laboratory platform.

Web Developer

11/2021 - 04/2022

D.D Vispute College of Science, Commerce & Management

Mumbai, India

- Developed and maintained the college website using HTML, CSS, JavaScript, and jQuery.

Web Developer

05/2021 - 11/2021

Alpha Laboratories

Mumbai, India

- Developed the company website using HTML, CSS, JavaScript, and jQuery.

Projects

CUTATO – Full-Stack Booking Platform

- Independently developing a full-stack booking platform for salons and barbers using Next.js, React, and JavaScript.
- Currently implementing responsive front-end interfaces, dynamic booking workflows, service selection, payment flow, and role-based dashboards.
- Building booking logic, dynamic scheduling, chatbot UI, and interactive components using structured React-based architecture.
- Using Git-based workflows and modular project structure for maintainable software development.

Robot Learning Pipeline for Household Robots (Ongoing)

- Research project under the LASR Group at TU Dresden focusing on robot learning for household environments.
- Working with literature and tools including RoboCasa, robomimic, OpenVLA, and SayCan to understand modern robotic learning pipelines.
- Investigating simulation, imitation learning, language-conditioned manipulation, and vision-language-action models for autonomous robots.

Using Formal Concept Lattices as Semantic Maps for Text Analysis

- Created semantic maps of language data using Formal Concept Analysis and graph-based reasoning.
- Applied Python-based implementation, structured data analysis, and visualization methods.

Uncovering the Dark Side of Large Language Models: Bias, Harm, and Toxicity

- Evaluated LLM behavior and analyzed conditions leading to toxic, biased, or discriminatory outputs.
- Used post-hoc explanation methods such as LIME and SHAP to explain model behavior.
- Implemented mitigation techniques such as reflection-based prompting in the LLM pipeline.

AgroSahayak

- Developed an AI/ML-powered agricultural platform for crop recommendation, disease prediction, and resource management.
- Integrated data-driven recommendations based on soil, weather, and crop-related parameters.

Positions of Responsibility

Communication Secretary <i>Student Council, Shah and Anchor Kutchhi Engineering College</i> – Coordinated between student teams and drafted permission letters, sponsorship documents, invitations, and MoUs.	2023 - 2024 <i>Mumbai, India</i>
Deputy Secretary <i>Research Cell, Shah and Anchor Kutchhi Engineering College</i> – Contributed to research-related projects and maintained the Research Cell website with regular updates.	2023 - 2024 <i>Mumbai, India</i>
Technical Lead <i>Research Cell, Shah and Anchor Kutchhi Engineering College</i> – Led technical project work and supported research and technical events.	2022 - 2023 <i>Mumbai, India</i>
Design Lead <i>Computer Society of India Student Chapter</i> – Created banners, certificates, digital backgrounds, and animations using Adobe tools, Figma, and Canva. Led a team of 10 members.	2022 - 2023 <i>Mumbai, India</i>

Licenses & Certifications

- **C Programming Examination** - IIT Bombay, Score: 93%
- **Data Structure and Algorithm** - ShapeAI
- **Python** - Developer Student Club
- **Data Visualization** - LetsUpgrade
- **Introduction to Machine Learning** - Infosys
- **Natural Language Processing** - GDG Ranchi

Language Skills

- **Mother tongue:** Marathi
- **Other languages:** English, Hindi
- **German:** Basic / A2, currently improving